

Design solid-state lithium-ion batteries with high thermomechanical stability with supervised learning simply by chemical formula

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Abstract

All-Solid-State Lithium-Ion Batteries (ASSLIBs) are critical for next-generation energy storage; however, material compatibility challenges persist, particularly in matching electrodes and electrolytes with compatible thermal expansion coefficients (TECs). Here, we introduce a machine learning approach for TEC prediction of lithium-containing compounds, requiring only chemical formulas as input. A baseline framework was developed using a Random Forest model trained on 428 experimental data points, further enhanced by a multi-level cascading algorithm and feature engineering. Validation through Density Functional Theory (DFT) calculations on LiF confirmed the model's reliability. The method was subsequently applied to screen ~100,000 materials in the Materials Project database, identifying candidates with TECs closely aligned with LiMn₂O₄ electrodes. This work bridges data-driven discovery and materials science, offering a

scalable solution to accelerate the design of thermomechanically stable lithium-ion batteries (LIBs). By integrating computational efficiency with physical insights, our approach enables smarter and faster material selection for advanced energy systems.

Key words: Thermal expansion coefficient, Solid-state electrolyte, Machine learning

1. Introduction

Solid-state batteries (SSBs) represent a promising solution for addressing the safety and stability issues associated with conventional lithium batteries¹. Unlike traditional liquid electrolytes, which are prone to combustion and expansion under extreme conditions, solid electrolytes are more stable. This advancement introduces critical challenges including poor solid-solid contact and thermal stress between the electrode and the electrolyte. The mismatch in TECs between the electrolyte and the electrode can result in two distinct failure modes: (1) excessive contraction reduces interfacial contact area, thereby increasing ionic transport resistance; or (2) excessive expansion generates interfacial cracks and thermal stresses, ultimately resulting in battery failure⁴. To address these questions, Bertrand et al. systematically investigated TECs of solid electrolytes and their compatibility with electrode

1 materials, establishing a foundation for optimizing SSB
2 performance through TEC alignment⁵.

3 Machine learning has become a transformative tool in materials
4 discovery, overcoming limitations of conventional screening
5 methods such as high costs and slow throughput^{6, 7}. Predictive
6 models enable rapid identification of materials with targeted
7 properties, significantly reducing development time and resources.
8 For instance, Kumar et al. demonstrated the feasibility of predicting
9 thermal expansion coefficients (TECs) using 271 structural features
10 and features generated from Magpie data, though sometimes
11 structural data acquisition remains a practical challenge⁸.

12 Chemical formula-based prediction has proven both effective and
13 critical for accelerating material discovery. Zijing Wan et
14 al. demonstrated that thermal stability can be predicted directly from
15 chemical formulas, significantly reducing computational costs and
16 enabling efficient preliminary screening⁹. Building on this approach, we
17 utilize chemical formulas to predict thermal expansion coefficients
18 (TECs), further extending the applicability of formula-driven
19 models in materials science

20 In this study, we propose an enhanced random forest framework
21 by integrating a multi-level cascaded architecture with residual

1 connections, effectively combining ensemble learning strategies. To
2 address data sparsity, chemical formulas were subjected to Centered
3 Log-Ratio (CLR) transformation, followed by the incorporation of
4 thermodynamic average electronegativity and descriptors derived
5 from Pymatgen's Magpie database. These steps collectively
6 improved the model's predictive capability for thermal expansion
7 coefficients (TECs). The optimized algorithm achieved a
8 determination coefficient (R^2) of 0.87, demonstrating robust
9 accuracy in TEC prediction and validating the efficacy of formula-
10 driven machine learning for materials screening.

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2. Method

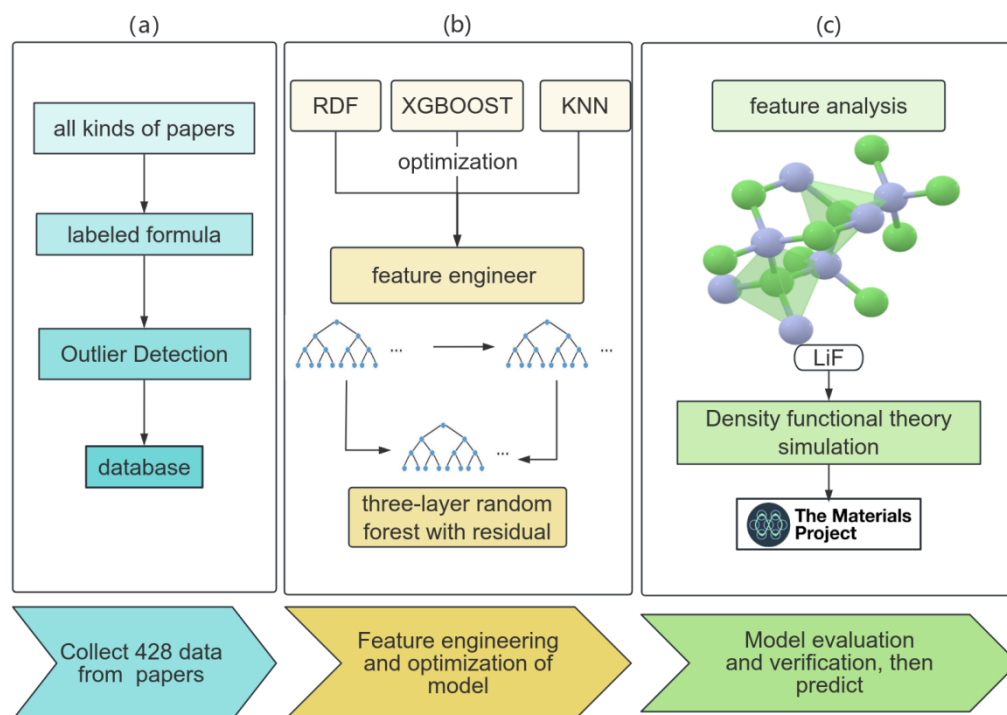


Figure 1 Research workflow (a) data collection process (b) model development including feature engineering and multi-layer cascaded structure (c) Validation & Application. The TEC screening process is outlined in Figure 1. First, data was collected from papers. Further, a model was selected and optimized via multi-layer architecture, residuals, and feature engineering. Finally, predictions were validated with DFT calculations to ensure reliability.

2.1 Database

A dataset of 428 compounds was compiled from experimental and simulated sources from all kinds of papers, as illustrated in Fig. 1 (a).

To ensure model robustness, outliers were systematically addressed by restricting the TEC range to $-20 \text{ K}^{-1} \leq \text{TEC} \leq 50 \text{ K}^{-1}$. Data points outside this range were excluded to mitigate bias from extreme values.

2.2 Model development

2.2.1 Model selection

Three machine learning models—Random Forest (RF), K-Nearest Neighbors (KNN), and eXtreme Gradient Boosting (XGBoost) were evaluated for TEC prediction with the elements' composition and quantities in chemical formulas serving as the feature matrix. Hyperparameter optimization was conducted via an exhaustive attack search, iteratively tuning individual parameters until convergence. This process identified parameter sets maximizing the coefficient of determination (R^2) on the test dataset. The model with the lowest MSE and the best R^2 of the three models on the test dataset will be chosen as the best model.

2.2.2 Feature Engineering

Feature engineering plays a pivotal role in bridging raw compositional data and machine learning models, particularly for sparse and high-dimensional material datasets. To address the inherent sparsity of elemental composition matrices and improve the accuracy, we prioritized feature construction to embed physicochemical significance. A key innovation is the incorporation of thermodynamic electronegativity (TEN). Electronegativity is particularly worthy of attention. McGuinness et al. demonstrated a strong correlation between electronegativity and TECs, establishing electronegativity as a descriptor for TEC prediction¹⁰. TEN,

1 Thermodynamic electronegativity (TEN), proposed by Tantardini and
2 Oganov et al. is a dimensionless metric that quantifies elemental
3 trends across the periodic table, explicitly capturing bond polarity and
4 ionic character. Its link to bond strength makes it closely related to
5 material thermal expansion coefficients¹¹.

6 The total TEN (EN_{sum}) of a compound is calculated as the
7 weighted sum of atomic electronegativities:

$$8 \quad EN_{\text{sum}} = \sum (EN_{\text{Atom}} \times N_{\text{Atoms}}) \quad (1)$$

9 Within this formula, EN_{Atom} : TEN of an atom; N_{Atoms} :
10 Stoichiometric count of the atom in the compound.

11 This formulation normalizes the chemical formula to quantify the
12 compound's overall TEN, serving as a key descriptor for predicting
13 TEC.

14 To enhance feature diversity, the Python library pymatgen's
15 Magpie was employed to generate descriptors, from which five
16 critical features were selected based on their R^2 value⁸. However,
17 matrices derived from elemental compositions inherently exhibit high
18 sparsity, leading to redundant predictions. To address this, Centered
19 Log-Ratio (CLR) transformation was applied, which not only
20 mitigates sparsity but also reduces data closure and spurious
21 correlations, thereby enhancing model stability. Additionally,

1 although random forests are inherently robust to feature scale
2 variations, significant disparities in feature magnitudes can distort
3 importance assessments, leading to degraded model performance¹².
4 To address this, feature scaling was implemented via an exhaustive
5 search method, optimizing parameters based on R^2 maximization.
6 This step ensures balanced contributions from heterogeneous features,
7 enhancing both model stability¹³.

$$8 \quad \text{CLR}(x_i) = \ln\left(\frac{x_i + \Delta}{G_x}\right) \quad (2)$$

9 **2.2.3 Ensemble Learning and Model Architecture**

10 Ensemble learning enhances model accuracy and robustness,
11 making it a cornerstone for practical applications. To further leverage
12 these advantages, we implemented a multi-level cascaded random
13 forest architecture inspired by Zhou et al's non-neural network deep
14 learning framework with cascaded forest structure which is effective
15 for small-scale datasets¹⁴. Each layer in the cascade iteratively refines
16 predictions from the preceding layer, progressively improving
17 performance with the upper layer's prediction as supplementary
18 features. Hyperparameters, including the optimal number of layers,
19 were systematically optimized through exhaustive search based on R^2
20 and confusion matrix. This approach enhances feature representation,

1 particularly effective for small-scale datasets. This approach enhances
2 feature representation and hierarchical learning, particularly effective
3 for small-scale datasets. In addition, residual connections were
4 introduced to enable random forests (RDFs) to iteratively learn the
5 residual errors between input and output, rather than directly mapping
6 raw features to predictions¹⁵. This simplifies the learning task, reduces
7 overfitting, and improves model generalizability.

8 **2.3 verification and application**

9 After configuring the model, as shown in Fig 1. (c), predictions
10 and validations were systematically executed. To assess universality,
11 DFT was initially performed on a simple structure (LiF). The
12 validated model was deployed to screen the Materials Project database,
13 identifying candidate materials with TEC compatible with LiMn_2O_4 .

14 **3. Results and discussion**

15 **3.1 model selection**

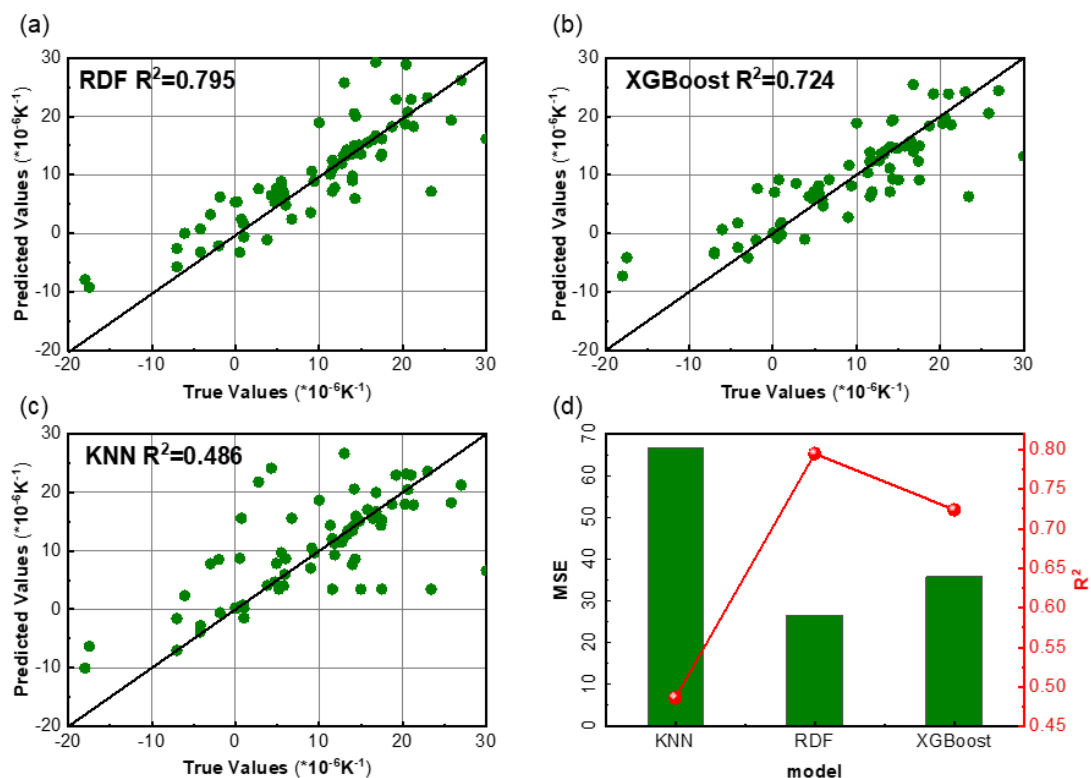


Figure2 Comparison chart of predicted and true values for three types of models on prediction set (a) KNN scatter plot of the prediction and true results on the test set (b) RDF scatter plot of the prediction and true results on the test set (c) XGBoost scatter plot of the prediction and true results on the test set (d) Statistical Chart of MSE and R^2 for Three Algorithms. The KNN, RDF, and XGBoost models are presented, comparing their predicted and actual values on the testing dataset. Each scatter point represents a dataset datum, with the vertical axis as predicted values and the horizontal axis as true values. The closer the scatter points lie along the $y=x$ diagonal, the better the model's prediction.

For the results of model selection, the comparative analysis of model performance is illustrated in Fig 2 (a - d). The RDF model's scatter plot points are closest to the $y=x$ diagonal, indicating the best predictive performance. Additionally, in Fig 3.(d), Quantitative validation further highlights RDF's dominance: it achieves the highest R^2 and the lowest mean squared error (MSE) among all candidates.

1 These metrics collectively solidify RDF as the optimal baseline model
 2 for subsequent analyses with R^2 of 0.79.

3 **3.2 Feature analysis**

4 Incorporating TEN as a descriptor improves the model's R^2 by 0.02,
 5 underscoring its significance in capturing bond polarity effects on
 6 thermal expansion. Subsequent feature selection from pymatgen's
 7 Magpie database, prioritized by R^2 contribution, identified five key
 8 descriptors (Table 1):

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Table 1 The features generated by Magpie after filtering

Feature	Abbreviation	Physical Property
thermodynamic_ electronegativity	TEN	Thermodynamic electronegativity is a scale based on bond dissociation energies and is used to describe an atom's ability to attract electron density.
MagpieData maximum NpValence	MNV	The maximum value of the number of valence electrons in the p orbital among all elements in the MagpieData database.
MagpieData minimum Electronegativity	MEN	The minimum value of electronegativity among all elements in MagpieData.
MagpieData mean CovalentRadius	MCR	The mean value of covalent radius of all elements in MagpieData.
MagpieData mode GSmagmom	MGS	The most frequent value of magnetic moments among all elements in MagpieData.
MagpieData avg_dev GSvolume_pa	ADGS	The average deviation of ground-state volume per atom (GSvolume_pa) among all elements in MagpieData.

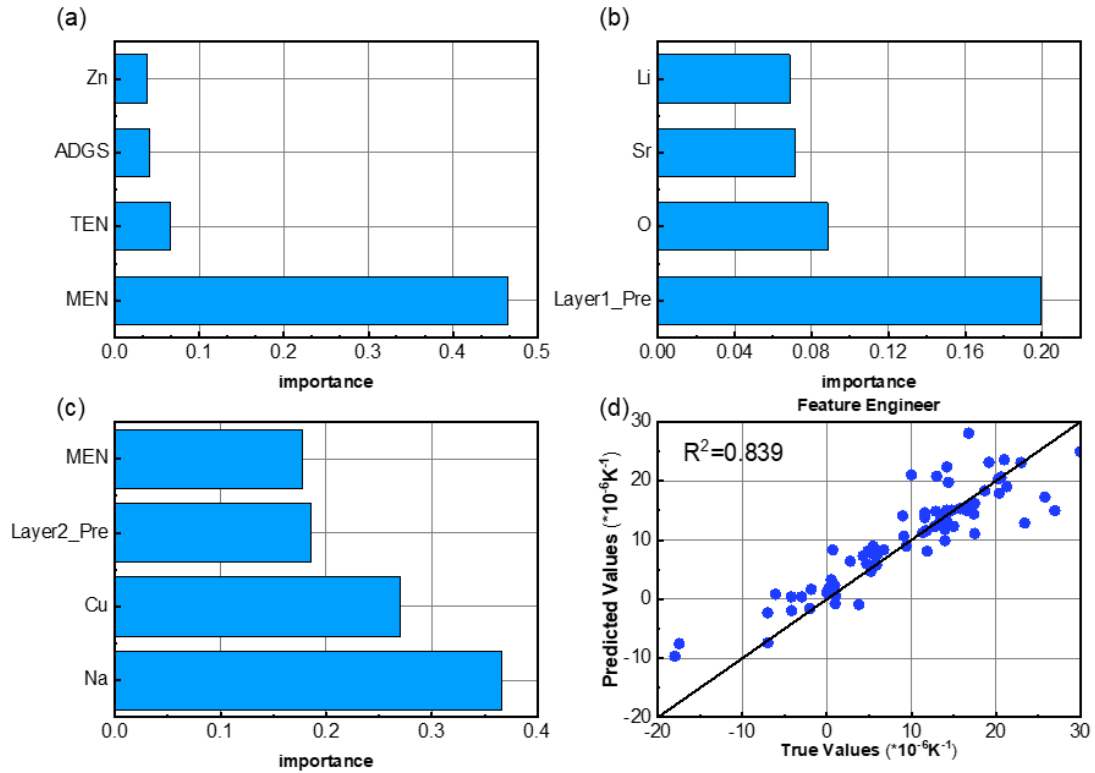


Figure 3 Feature importance comparison chart and final results (a) feature importance comparison chart of layer1 (only the first four large ones) (b) feature importance comparison chart of layer2 (only the first four large ones) (c) feature importance comparison chart of layer2 (only the first four large ones) (d) final RDF algorithm scatter plot of the prediction and true results on the testing set and each decision tree in a random forest splits nodes to minimize the variance of the target variable. When a feature is used for splitting, the resulting variance reduction is calculated. Summing this variance reduction across all trees and dividing by the number of trees yields the feature's importance.

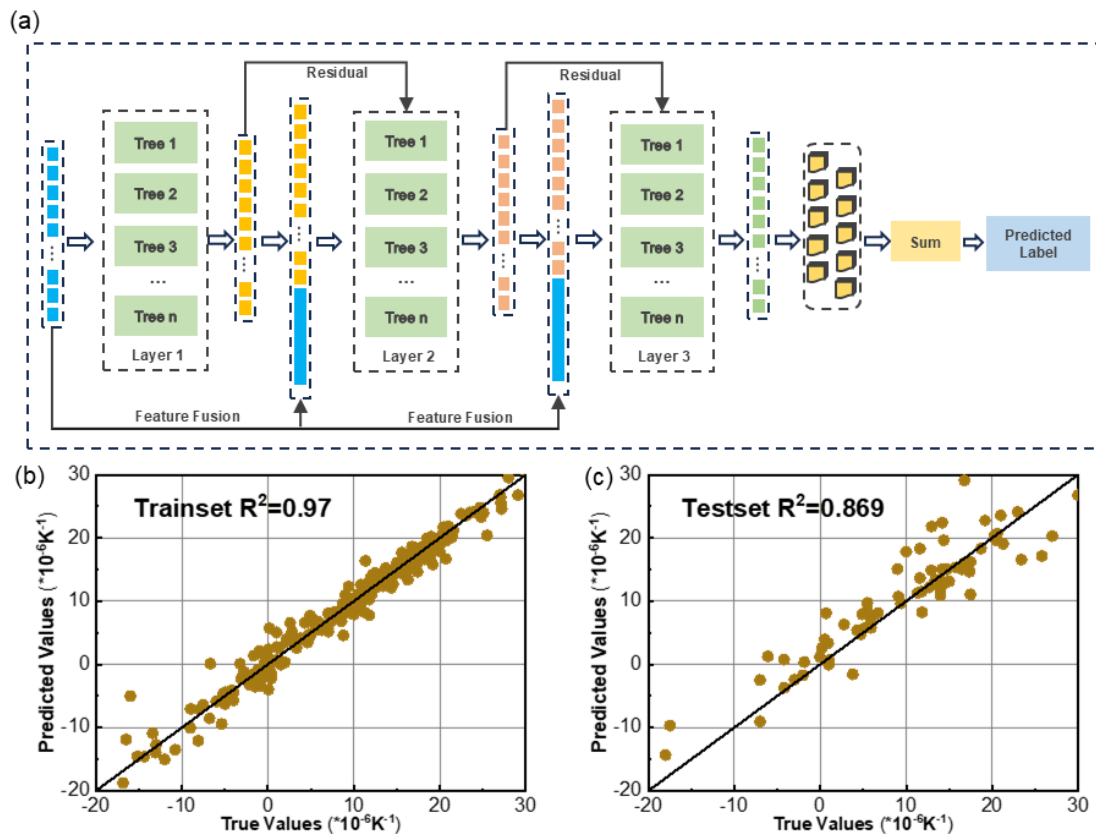
RDF evaluates feature importance by aggregating contributions across all decision trees. As shown in Fig 3 (a), TEN and minimum electronegativity (MEN) dominate the first layer, confirming electronegativity's critical role in TECs. In Fig 3 (b-c), the high importance of the first and second layers' predictions indicates that feature fusion through residual connections is beneficial. Moreover, the second layer demonstrates a more balanced distribution of

1 importance, utilizing additional features not employed in the first
2 layer, thereby enhancing information efficiency. In conclusion, the
3 importance distribution confirms that all our optimization operations
4 are justified. Additionally, Fig 3 (d) met expectations, achieving a R^2
5 score of 0.839.

6 **3.3 Ensemble Learning and Model Architecture**

7 During optimization of the multi-level cascaded RDF architecture,
8 Fig. 4 (b) demonstrates near-perfect alignment between predicted and
9 experimental TECs in the training set, suggesting potential overfitting
10 and prompting termination of additional layers. Additionally, Fig 4 (c)
11 reveals strong generalization on the test set, with predictions closely
12 matching true values and an R^2 exceeding 0.85, confirming the
13 model's suitability for small-scale datasets. Finally, the final algorithm
14 flowchart is shown in Fig 4 (a). First, in the first layer input features
15 generate initial predictions. After that, the residual, the difference
16 between initial predicted values and true values, serves as the second
17 layer's target. The first layer's predicted values, directly or after
18 transformation, are concatenated with original features to form the
19 second layer's input feature matrix. This process repeats until the third
20 level. The final prediction is the sum of all layers' predictions. This

1 hierarchical method enhances predictions by iteratively refining
2 residuals and enriching features. In conclusion, a reasonably good fit
3 has been achieved with the R^2 of 0.87 shown in Fig 4 (c).



8 Figure 4 three-layer residual RDF algorithm flowchart and the model results (a) three-layer residual
9 RDF algorithm flowchart (b) three-layer residual RDF algorithm scatter plot of the prediction and
10 true results on the test set (c) three-layer residual RDF algorithm scatter plot of the prediction and
11 true results on the training set. In fig 4.(a), the blue square represents the original feature values, and
12 each layer is trained to fit the residuals of the previous layer. Finally, add them up as the solution.

15 3.4Verification

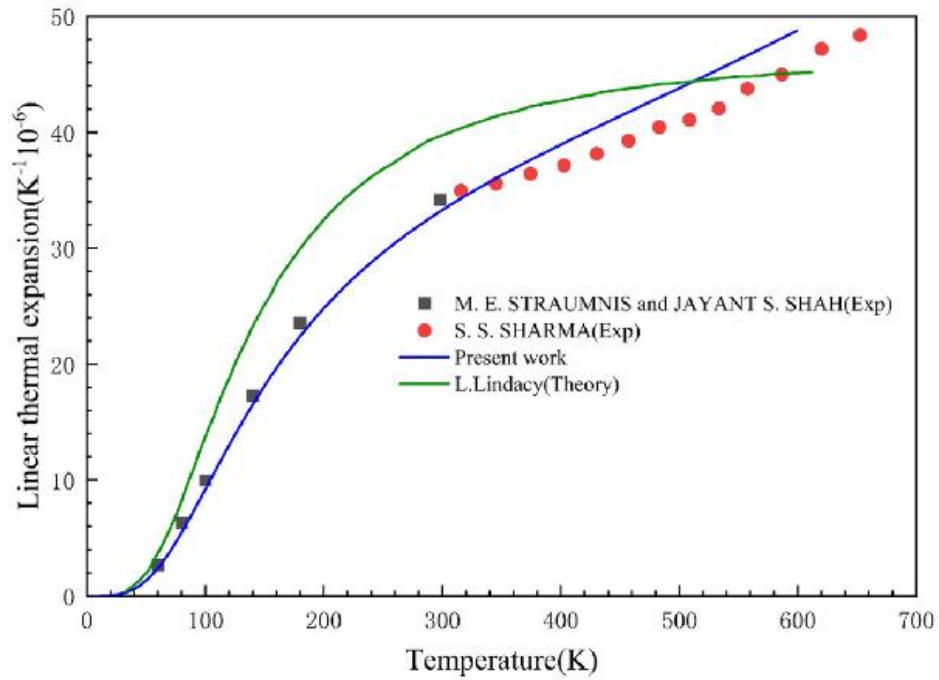


Figure5 The scientific simulation of the thermal expansion coefficient of LiF, the blue line is the validation value

To validate the reliability of the model, Density Functional Theory (DFT) calculation was performed on LiF—a typical compound in battery. As shown in Fig 5, the model predicts a thermal expansion coefficient (TEC) of 30.8 K^{-1} for LiF, closely aligning with the experimental value of 34 K^{-1} (deviation: 9.4%). This minor discrepancy likely stems from the exclusion of structural parameters such as lattice symmetry and atomic coordination in our formula-driven approach. Consequently, while the model demonstrates robust accuracy for magnitude-level screening of electrode-electrolyte pairs, its precision remains constrained for atomic-level design requiring explicit structural considerations.

3.2 Prediction

Following model construction and validation, we systematically screened lithium-containing compounds from the Materials Project database with chemical formula information and without elements not present in the training data. After screening, using the trained model, the thermal expansion coefficients for all 8,271 compounds are predicted. It is found that pure metal compounds may have poor estimates because there are not many pure metal compounds in the training dataset. Candidate materials with compatible thermal properties with common anode and cathode are identified and collected followed.

4. Conclusion

The basic random forest was optimized by forming a three-tiered cascading random forest with residuals. Subsequently, feature construction was performed on chemical formulas to enhance the model's determination rate. But our model can just predict the thermal expansion coefficient within a certain range because the thermal expansion coefficient is always strongly related to the structure and for the same chemical formula there may be completely different thermal expansion coefficients. So further verification is necessary.

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19 Table 2 List of cathode-electrolyte materials with thermomechanical compatibility according to
20 ML-predicted CTE values.

Cathode	Electrolyte	Difference $\times 10^{-6}$ (K^{-1})	Average $\times 10^{-6}$ (K^{-1})
LiMn ₂ O ₄	LiPr(GeO ₃) ₄	0.57856	7.72975
C	LiIn ₄ (Ge ₂ O ₇) ₂	0.30536	1.15268
LiCoO ₂	LiNdTiO ₄	0.30037	12.84981
LiFePo ₄	LiNdTiO ₄	0.49963	12.44981

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